Milk can be treated at 141°C for 2 seconds (This method employ temperature above 100°C).

b) Temperature at 100°C:

i) Water at 100°C (Boiling):

Boiling is one of the moist heat sterilization methods. It kills vegetative forms of bacterial pathogens, almost all viruses and fungi (including their spores) within 10 minutes, usually much faster.

- Most vegetative bacteria will die in 5-10 minutes when immersed in boiling water, but some spores will survive at this temperature for several hours.
- Articles sterilized by this method cannot be stored for a long time.

ii) Steaming at 100°C (Tyndallization):

It is a process discovered by John Tyndall in 19th century for sterilizing substances to kill the spores of bacteria. The process of exposure of materials to steam at 100°C for 20 min for three consecutive days is known as tyndallization. First exposure kills all the vegetative forms and in the intervals between heating, the remaining spores germinate into vegetative forms which are killed on subsequent heating. Tyndallization is also called fractional sterilization or intermittent boiling.

c) Temperature above 100°C:

Moist heat sterilization can be carried out at temperature above 100°C in order to destroy bacterial endospores. This requires the use of saturated steam under pressure. This is achieved using autoclave.

Autoclave

Sterilization using an autoclave is most effective when the organisms are either contacted by the steam directly or contained in a small volume of aqueous liquid (primarily water). The temperature used in autoclave is 121°C at 15 lbs (pounds) pressure for 15 minutes (Figure 4. 2a & b).

Autoclaving is used in sterilizing culture media, instruments, dressings, applicators, solutions, syringes, transfusion equipment, pharmaceutical products, aqueous solutions and numerous other items that can withstand high temperatures and pressures. The same principle of autoclaving applies for the common household pressure cooker used for cooking food.

Factors influencing sterilization by heat:

Sterilization by heat depends upon various factors such as time, temperature employed, number of microorganisms, spores and nature of material to be sterilized.

Quality control of moist heat sterilization:

To check the efficiency of moist heat sterilization, the indicator commonly used is the paper strips containing spores of *Bacillus stearothermophilus.*

4.5 Radiation

Radiation is commonly employed for sterilizing heat sensitive materials such as disposable plastic products and materials that cannot withstand moisture.

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The most effective type of radiation to sterilize or reduce the microbial burden in the substance is through the use of electromagnetic radiations. Figure 4.3 shows different types of electromagnetic radiations. Radiation has various effects on cells, depending on its wavelength, intensity and duration of explosure (Flowchart 4.2). Radiation that kills microorganism is of two types namely ionizing and nonionizing.

a) Non-ionizing radiation

Infra-red rays and ultra-violet rays are non ionizing radiation.

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